

Martim Vicente

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Education

Instituto Superior Técnico

BSc in Computer Science and Engineering

Lisbon, Portugal

Sep. 2024 – Jun. 2027

- **Average:** 15.75 / 20.0
- **Honors:** Técnico ALUMNI Scholarship, Gulbenkian Merit Scholarship, WARD Analytics Scholarship

Skills

Languages: C++, C, Python, Go, RISC-V Assembly

Tools: Linux, Git, CMake, Make, Docker, QEMU, Renode, MQTT

Embedded: STM32, ESP32, FreeRTOS, CMSIS-RTOS2, UART, I2C, SPI, ADC, PWM, GPIO, CAN

Testing: Pytest, GoogleTest, GitHub Actions

Experience

Software Team Lead, Rocket Experiment Division – Lisbon, PT

May 2026 – Present

- Led software development and integration for a 6-person team, owning software architecture, engineering practices, and cross-system integration.
- Architected, developed, and validated firmware for ignition, fluid filling, and avionics systems, implementing concurrent task execution, inter-task communication, and fail-safe control.
- Implemented drivers for IMUs, GPS, electrovalves, servos, and sensors within a modular, hardware-agnostic C HAL, enabling hardware portability and mock-based unit testing.
- Rebuilt the GSE backend in asynchronous Python using MQTT, reducing production code by 47% and introducing a telemetry simulator for hardware-independent bench testing and recorded-data replay, enabling end-to-end validation from telemetry generation through the gateway to dashboards at up to 100 Hz.
- Migrated flight and ground systems from an in-house telemetry protocol to MAVLink v2, standardizing message formats across all systems.

Embedded Software Engineer, Rocket Experiment Division – Lisbon, PT

Sep. 2025 – May 2026

- Designed and implemented the initial firmware for the rocket's fluid filling system, developing control logic, hardware interfaces, and safety-critical state handling.
- Built a centralized technical documentation portal covering software architecture, hardware interfaces, and testing procedures, establishing a shared reference for development and integration.

Software Engineer, Técnico Investment Club – Lisbon, PT

Sep. 2025 – May 2026

- Re-engineered the Python trading runtime into decoupled ingestion and strategy services, improving fault isolation, modularity, and maintainability.
- Designed the technical architecture and requirements for a low-latency execution engine, defining asynchronous event processing, order validation, and exchange interfaces.